Complex Analysis Homework 3

1. The *Jacobian* of a mapping $u = u(x, y)$, $v = v(x, y)$ from the xy-plane to the uv-plane is defined to be the determinant

$$J(x_0, y_0) := \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix},$$

where the partial derivatives are all evaluated at (x_0, y_0).

(a) Show that if $f = u + iv$ is analytic at $z_0 = x_0 + iy_0$, then $J(x_0, y_0) = |f'(z_0)|^2$.

(b) Let $f(z)$ be a bounded analytic function, defined on a bounded domain D in the complex plane, and suppose $f(z)$ is one-to-one. Show that the area of $f(D)$ is given by

$$\text{Area}(f(D)) = \iint_D |f'(z)|^2 \, dxdy.$$

(c) Sketch the image of the circle $\{|z-1| \le 1\}$ under the map $f(z) = z^2$. Compute the area of the image.

(d) Compute

$$\iint_D |f'(z)|^2 \, dxdy,$$

for $f(z) = z^2$ and D the open unit disk $\{|z| \le 1\}$. Interpret your answer in terms of areas.

2(a) Evaluate $(-1+i)^i$, i^i and $(-1-i)^i$.

(b) As we all know, $(-1+i)\,i = -1-i$. Is $\left(\text{principal value of } (-1+i)^i\right) \times \left(\text{principal value of } i^i\right)$ $= \left(\text{principal value of } (-1-i)^i\right)$ correct?

(c) Is there any equality in $(-1+i)^{2i}, \left((-1+i)^2\right)^i, (-1+i)^i \times (-1+i)^i$?

3. Solve $\sin z = \dfrac{1}{\sin 2\theta}$ where $0 < \theta < \frac{\pi}{2}$.

4. Solve $z^4 - (5-14i)z^2 - 2(5i+12) = 0.$

5. Let $S_1 = \{z \in \boldsymbol{C} : |z| < 1\}$ and $S_2 = \{z \in \boldsymbol{C} : \operatorname{Re} z < 0\}$. Define a function $f : S_1 \to S_2$ by $f(z) = \dfrac{z+i}{z-i}$.

(a) Show that $\operatorname{Re}(f(z)) < 0$ for all $z \in S_1$.

(b) Show that f is injective and surjective.

(c) Express the inverse function of f.